Trent Haines

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Experience

Roblox — **Feb 2024 – Present**

Software Engineer - Backend, Payment Systems — *San Mateo, California*

- Traced the full service communication network for Payment Search, revealing redundant paths and fragmented data sources. Rearchitected Payment Search to directly communicate with a new Elasticsearch-backed API, eliminating middlemen and establishing a single source of truth.
- Redesigned the internal representation of 'product', replacing an inflexible 6-table relational schema to unified single-table with Protobuf-structured data
- Delivered a zero-downtime migration of 10B+ core billing entity records. Led end-to-end data validation, created interface adapters, and refactored legacy systems to maintain seamless compatibility.
- Built a drop-in dual-read/dual-write framework adopted as the organization standard across 10 engineering teams. Abstracted away cutover complexity through configurable execution modes, background orchestration, error handling, rollout controls, mismatch telemetry, automatic alerting.
- Evolved subscription renewal management from an unreliable table-scan processor to a workflow-driven Temporal orchestrator, improving availability from <99% to >99.99%.

Amazon — **May 2022 – August 2022**

Software Engineering Intern - Fullstack, Route Planning Optimization — *Austin, Texas*

- Developed an interactive web app enabling Amazon science teams to analyze delivery route execution patterns, identifying 14 operational metrics and implementing end-to-end data pipeline from database queries to frontend visualizations with side-by-side route comparisons, guiding route-planning optimization decisions.

FireEye — **May 2021 – August 2021**

Software Engineering Intern - Backend, Telemetry — *Dallas, Texas*

- Improved the threat detection engine's reliability by migrating container infrastructure from ECS to EKS and introducing monitoring through Grafana dashboards and PromQL metrics to track container performance, throttling events, and resource utilization.

University of Texas at Dallas — **May 2016 – August 2017**

NanoTechnology Research Intern — *Richardson, Texas*

- Developed applications for thermally actuated nylon artificial muscles, with prototypes incorporated into a successful NASA STTR Phase II grant application.
- Synthesized organic aerogels and conducted experiments to test and optimize their material properties.

Education

University of Texas at Dallas — **August 2019 - May 2023**

Master's + Bachelor's of Science in Computer Science — *Richardson, Texas*

- 4-Year Fast-Track MS/BS
- Full Ride Merit Scholarship
- Computing Scholar's Honors

Relevant Coursework

- Algorithm Analysis
- Statistical Methods for AI/ML
- Distributed Computing
- Database Systems
- C++ in UNIX Environment
- Operating Systems

Projects

Convex Hull Algorithm Visualizer | *JavaScript, HTML, CSS* — **April 2022**

- Created a react application that visualizes three different Convex Hull algorithms to help Computational Geometry students better understand and compare them. Students are able to place points down to create custom data-sets to better understand the strengths of the various algorithms in different scenarios

Blind Accessibility Chrome Extension | *JavaScript, Python* — **April 2020**

- Collaborated with a team to create a Chrome extension using Google's Cloud Vision API to provide text labeling for images on webpage. These labels can be read aloud using text-to-speech, or can assist in image captioning

Technical Skills

Languages: C#, C++, Python, Java/Kotlin, JavaScript/TSX
Frameworks and Libraries: ASP.NET Core, gRPC/Protobuf, Temporal Workflows
Tools: Git, Linux, Docker, Grafana